

Yiyang Ge

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EDUCATION

University of Washington Seattle, WA	Sept 2026 - Jun 2028(<i>expected</i>)
MS in Mechanical Engineering	
Zhejiang University Hangzhou, China	Sept 2022 - Jun 2026
BEng in Automation	GPA:4.13/4.3
◦ Relevant Courses: Principles of Automatic Control(95), Air Robots(94), Robot Modeling and Control(93)	
Cornell University Ithaca, NY	Jan 2025 - May 2025
Exchange Student in Electrical and Computer Engineering	GPA:4.15/4.3
◦ Relevant Courses: Intro to Reinforcement Learning(A+), Robot Learning(A)	

RESEARCH INTERESTS

Dexterous Manipulation, Reinforcement Learning, Imitation Learning, Visuotactile Perception.

PUBLICATION

Yiyang Ge, Sihan Bai, Ke Jin, Binghen Zhang, Siyun Wang, Qi Ye. "Vision-Initialized Manipulation with Image-Aligned Proprio-Tactile Feedback". Submitted to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. Under review.

Jiaying Chen, Bingheng Zhang, Qingtao Liu, **Yiyang Ge**, Wenzhe Ouyang, Anjun Chen, and Qi Ye. "Relational Dexterity: Transferring Human Dexterity Via Relational Geometric Prior" Accepted to the *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

RESEARCH EXPERIENCE

Vision-Initialized Manipulation with Image-Aligned Proprio-Tactile Feedback	Jun 2026 - Sept 2026
Zhejiang University, PerPLe Lab	Hangzhou, China
◦ Motivated by human manipulation with limited visual feedback, studies Vision-Initialized Manipulation setting where policy uses a single initial RGB image and subsequently operates using only tactile and proprioceptive feedback.	
◦ Projected fingertip positions and tactile measurements into the initial camera frame to construct image-aligned fingertip-depth and tactile-saliency representations, enabling fusion of temporally mismatched sensory inputs.	
◦ Designed an auxiliary current-image reconstruction objective, weighted by temporally aggregated fingertip occupancy , to emphasize interaction-relevant regions during representation learning.	
◦ Evaluated the method on four simulated manipulation tasks and a real-world bottle-liftup task, achieving a 29% relative improvement in mean success rate over imitation-learning baselines under the same visual constraint and approaching the performance of policies with updated vision on selected tasks.	
Automated Reward Modeling for Real-World Robot Reinforcement Learning	Jan 2026 - May 2026
UC Davis, DARE Lab & Zhejiang University, PerPLe Lab	Davis, CA
◦ Investigated sample-efficient offline robot reinforcement learning using learned visual reward functions.	
◦ Fine-tuned a pretrained Valine-Implicit Pretraining(VIP) model with a small set of demonstrations to estimate task rewards.	
◦ Integrated the learned reward model with Reward-Weighted Regression (RWR) to enable offline policy improvement without manually designed rewards.	
◦ Evaluated the approach in ManiSkill and deployed it on an AgileX Piper robot arm for a real-world cube pick-and-place task.	
Transferring Human Dexterity via Relational Geometric Priors	Oct 2025 - Jan 2026
Zhejiang University, PerPLe Lab	Hangzhou, China

- Investigated retargeting human hand motion-capture data into task-feasible trajectories for dexterous robot hands by reinforcement learning.
- Reproduced and evaluated the ManipTrans baseline, conducted comparative experiments, and contributed to method refinement.

PROJECTS

Spine-Enabled Quadruped Locomotion in Simulation

Jan 2026 – Feb 2026

- Modified the Unitree Go2 URDF by adding an actuated spinal joint to investigate spine-assisted locomotion.
- Reproduced the Unitree RL Gym pipeline and trained a reinforcement learning policy in Isaac Gym for turning through narrow passages.
- Achieved stable sim-to-sim transfer from Isaac Gym to MuJoCo through domain randomization and alignment of actuator dynamics, PD control parameters, action scaling, and joint limits.

Fingertip Force Control of an Under-actuated Exoskeleton

Aug 2024 – Dec 2024

- Modeled the kinematics of hand exoskeleton and developed software to compute fingertip pose in real time.
- Redesigned the mechanical structure to integrate a BLDC motor, and implemented force control using a Field-Oriented Control (FOC) framework with a PID controller.
- Achieved real time force feedback with an average torque error of approximately 0.5 Nm.

Cascaded PID Controller for a Quadcopter

Jul 2024 – Aug 2024

- Designed a cascaded PID controller for a quadcopter, implemented it with a ESP32 microcontroller.
- Designed and assembled a custom PCB flight controller, integrating sensors, microcontrollers, and communication modules.

TECHNOLOGIES

Computer Skills: Python, Pytorch, Git, Numpy, C, Matlab, Linux

Hardware Skills: basic PCB design, robotics & embedded system development

AWARDS

Outstanding Graduate, Zhejiang University	<i>2026</i>
Second Prize Scholarship of Zhejiang University	<i>2023, 2025</i>
First Prize in Zhejiang University Robot Competition (Transport Vehicle Track)	<i>2024</i>
Second Prize of Theoretical Physics Competition for University Students in Zhejiang Province	<i>2023</i>